

# Fitness Tracker Data Analysis Report



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## Business Objective

The business objective is to analyze smart fitness tracker data to understand user behavior patterns (activity, sleep, calories, BMI) and help improve the marketing strategy.

These insights will help Strava improve its marketing strategy and encourage healthier lifestyles among users. In simple words: We want to understand how people use their fitness trackers and whether they follow healthy habits.

## Dataset Description

The dataset contained 18 files collected from fitness tracking device. For our analysis we used four datasets namely

- daily activity records
- activity intensity records
- sleep tracking logs
- Weight and BMI records

## Methodology

1. Data Cleaning – handled missing values, standardized date formats, removed duplicates.
2. Data Integration – merged multiple datasets into a single master dataset.
3. Feature Engineering – created derived metrics such as sleep efficiency.
4. Exploratory Data Analysis – used visualizations to understand behavioural patterns.
5. Insight Generation – identified trends in activity, sleep, and BMI.

# Exploratory Data Analysis and Business Insights

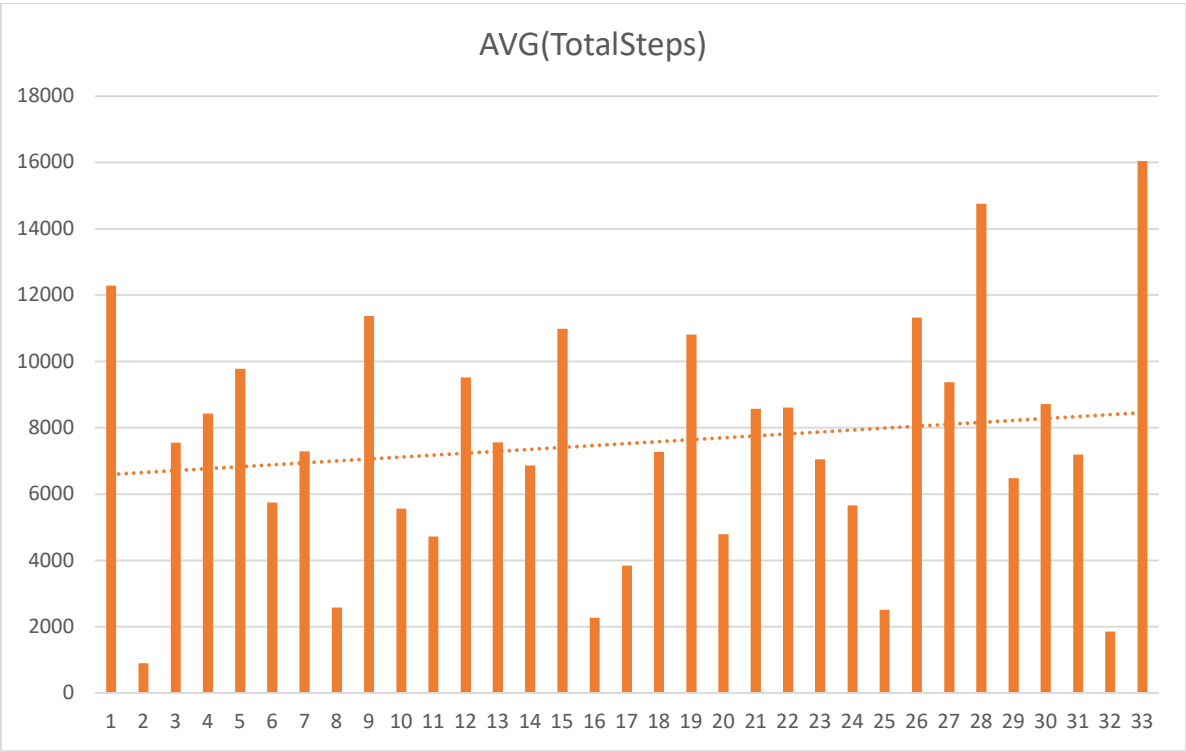
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## User Engagement Insights

### 1. Average User Activity

Users walk an average of ~7k–8k steps per day, which indicates moderate activity levels but still below the recommended 10,000 steps benchmark.

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## 2. Most Active Users

A small group of users contributes to the highest step counts, indicating a gap between highly active users and sedentary users.

Business implication:

Fitness apps can target low-activity users with personalized reminders and challenges.

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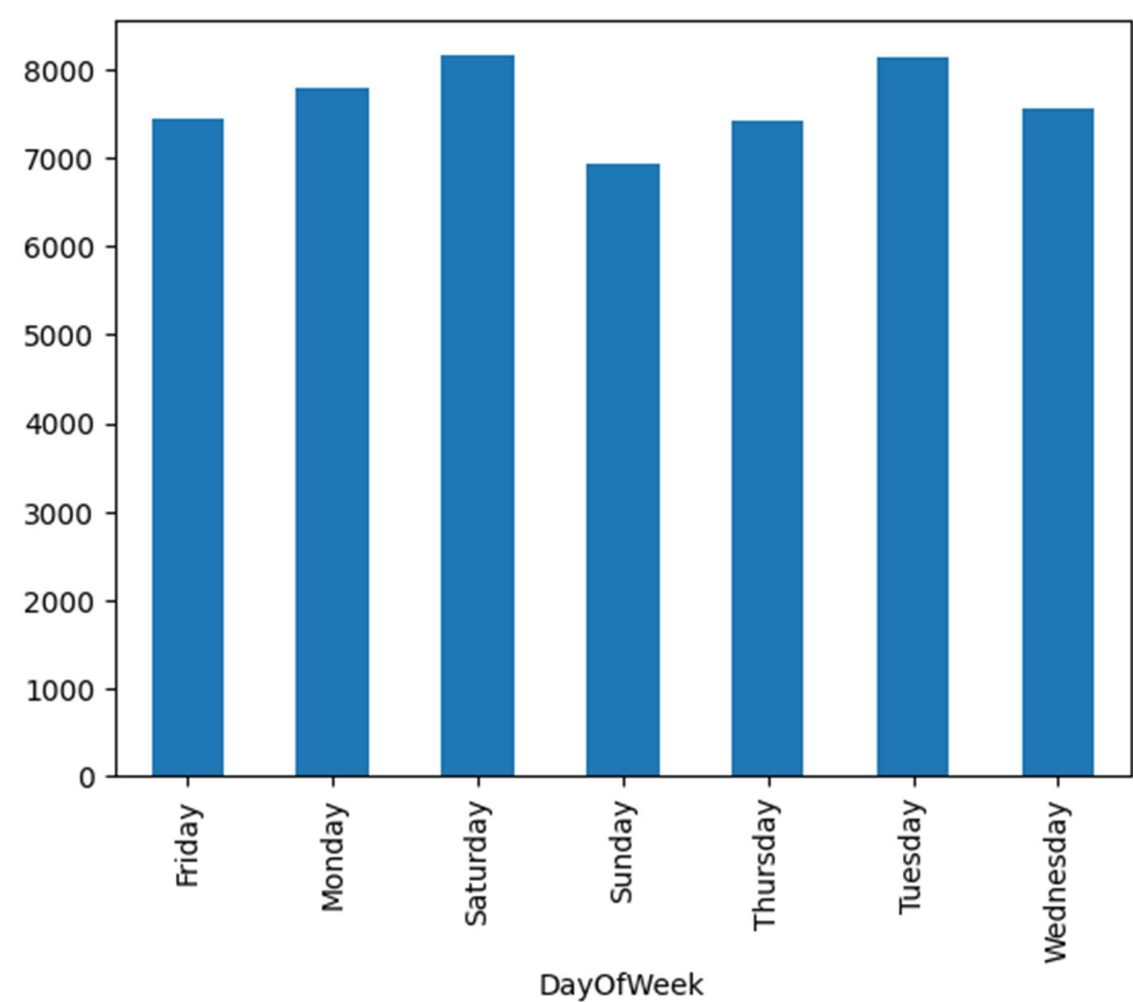
| Id         | AVG(TotalSteps) |
|------------|-----------------|
| 8877689391 | 16040           |
| 8053475328 | 14763           |
| 1503960366 | 12286           |
| 2022484408 | 11371           |
| 7007744171 | 11323           |
| 3977333714 | 10985           |
| 4388161847 | 10814           |

### 3. Activity Pattern

Users tend to be more active on weekdays compared to weekends.

Possible reason:

Work-related movement or structured routines during weekdays.



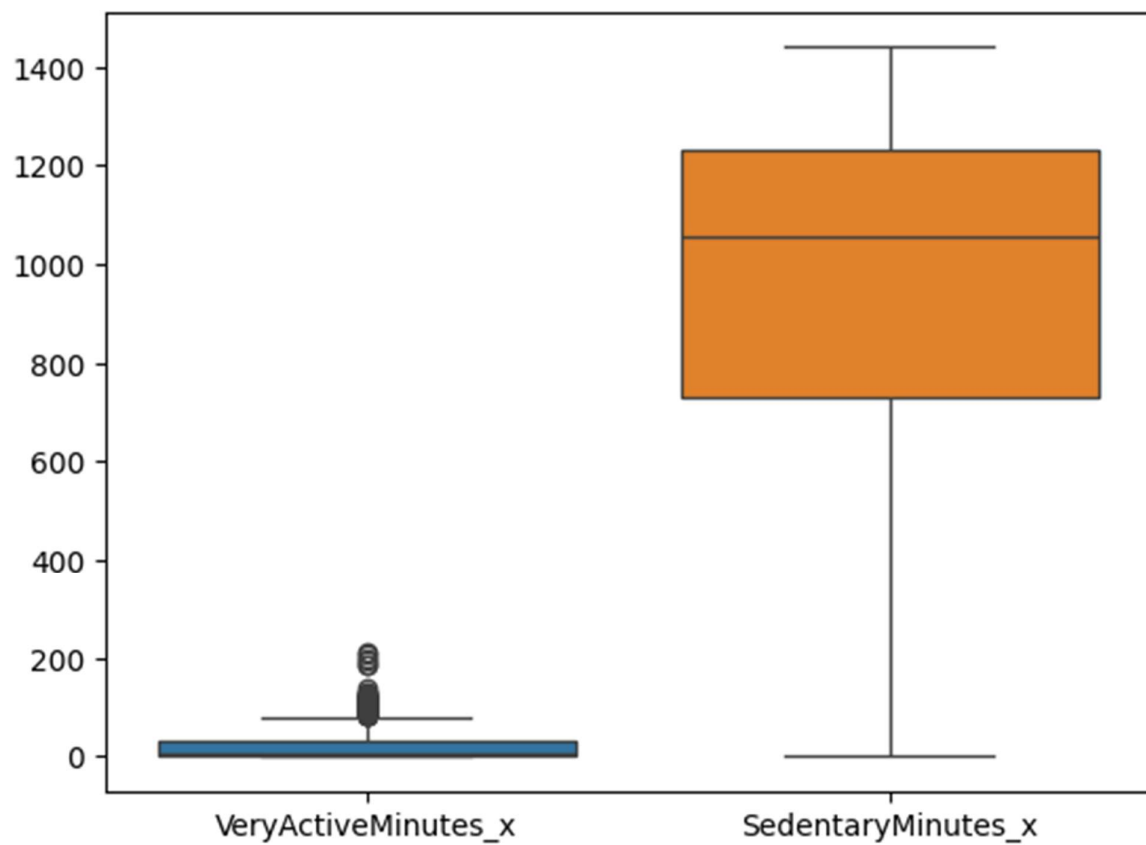
## 4. Sedentary Behavior

Sedentary minutes are significantly higher than active minutes for most users.

Business implication:

The app should introduce standing reminders or micro-activity notifications.

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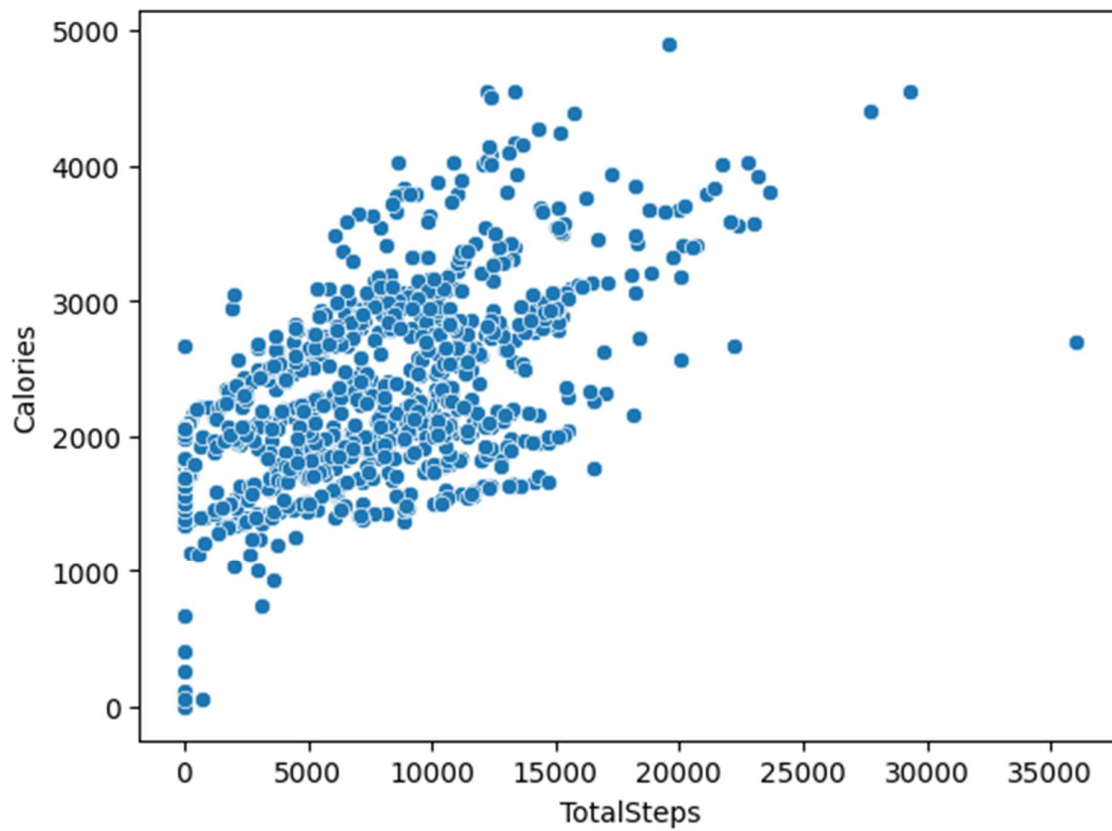
## 5. Steps vs Calories

There is a strong positive correlation between steps taken and calories burned.

Insight:

Increasing daily step count directly improves calorie expenditure.

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# Sleep Pattern Insights

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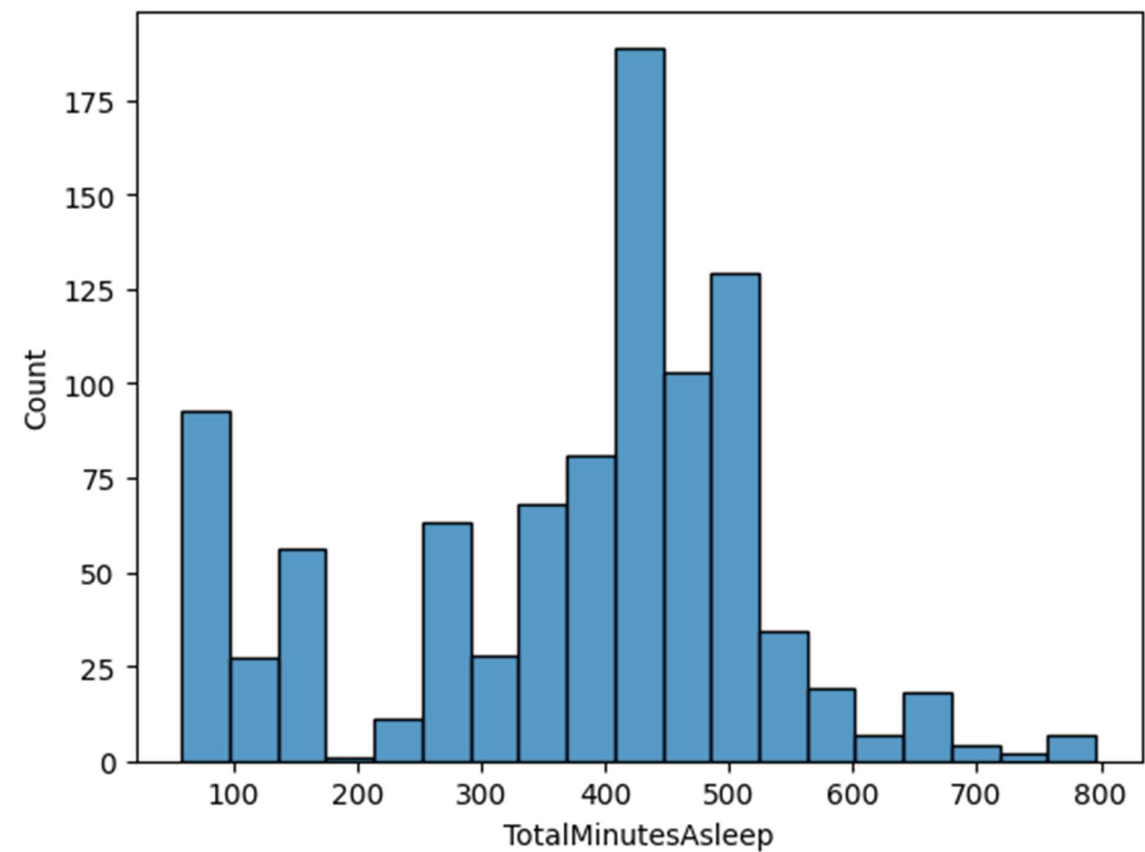
## 6. Average Sleep Duration

Average sleep duration ranges around 6–7 hours per night.

Insight:

Many users sleep slightly below the recommended 7–8 hours.

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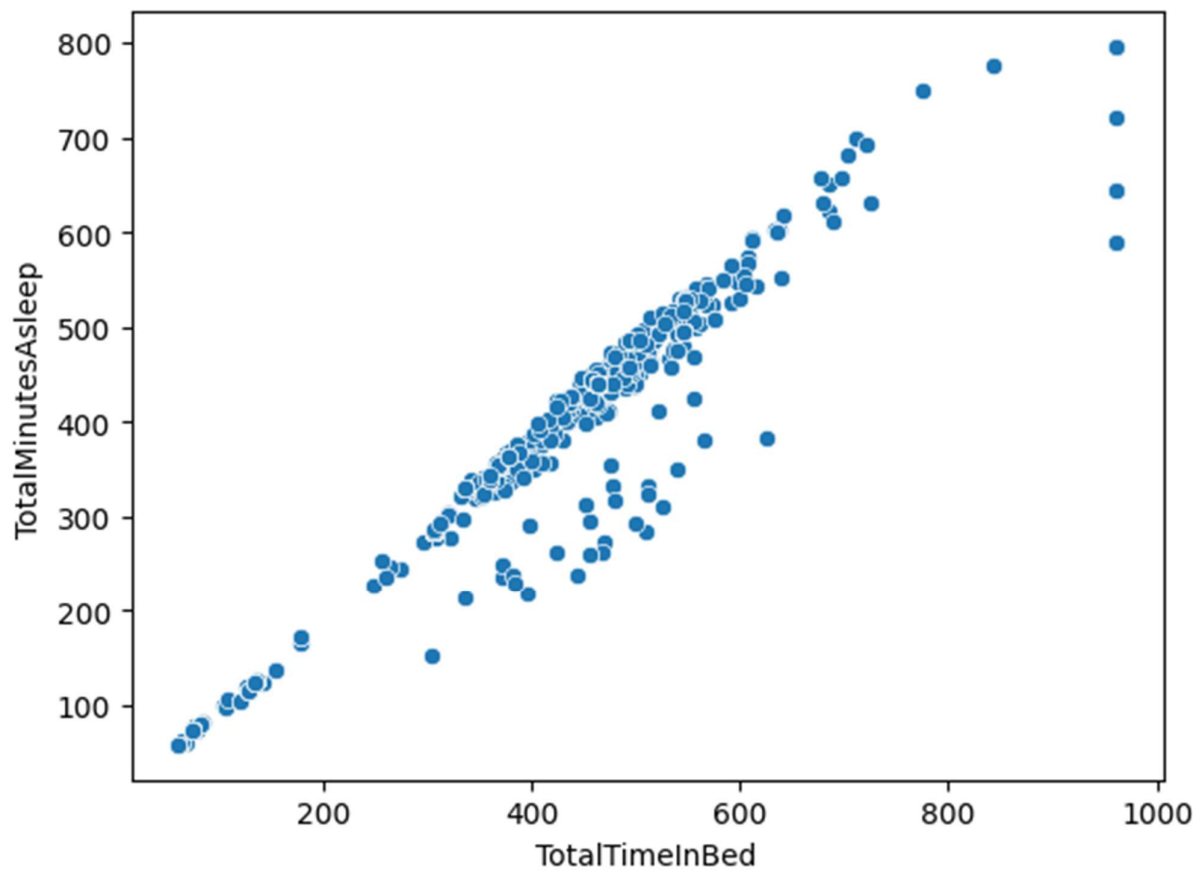


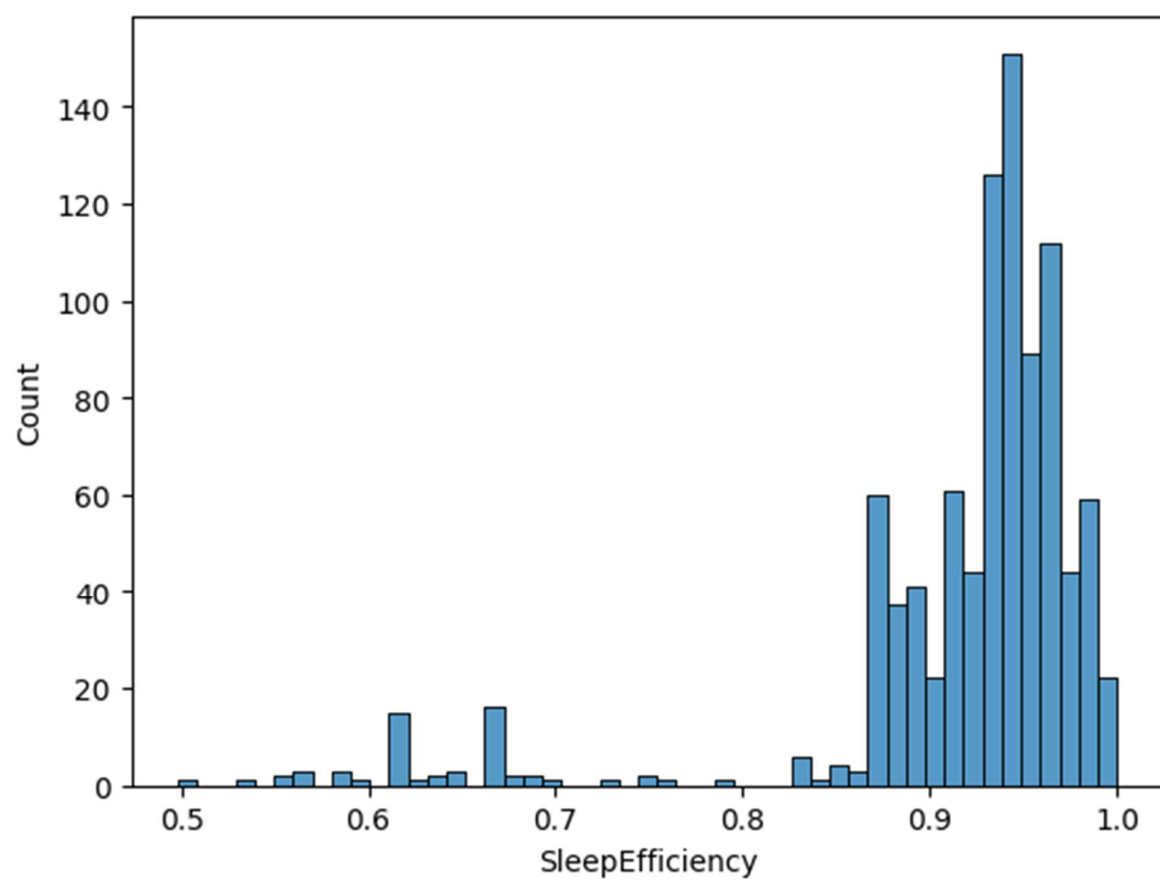


## 7. Sleep Efficiency

Most users show high sleep efficiency (80–90%), meaning the time spent in bed is mostly spent sleeping.

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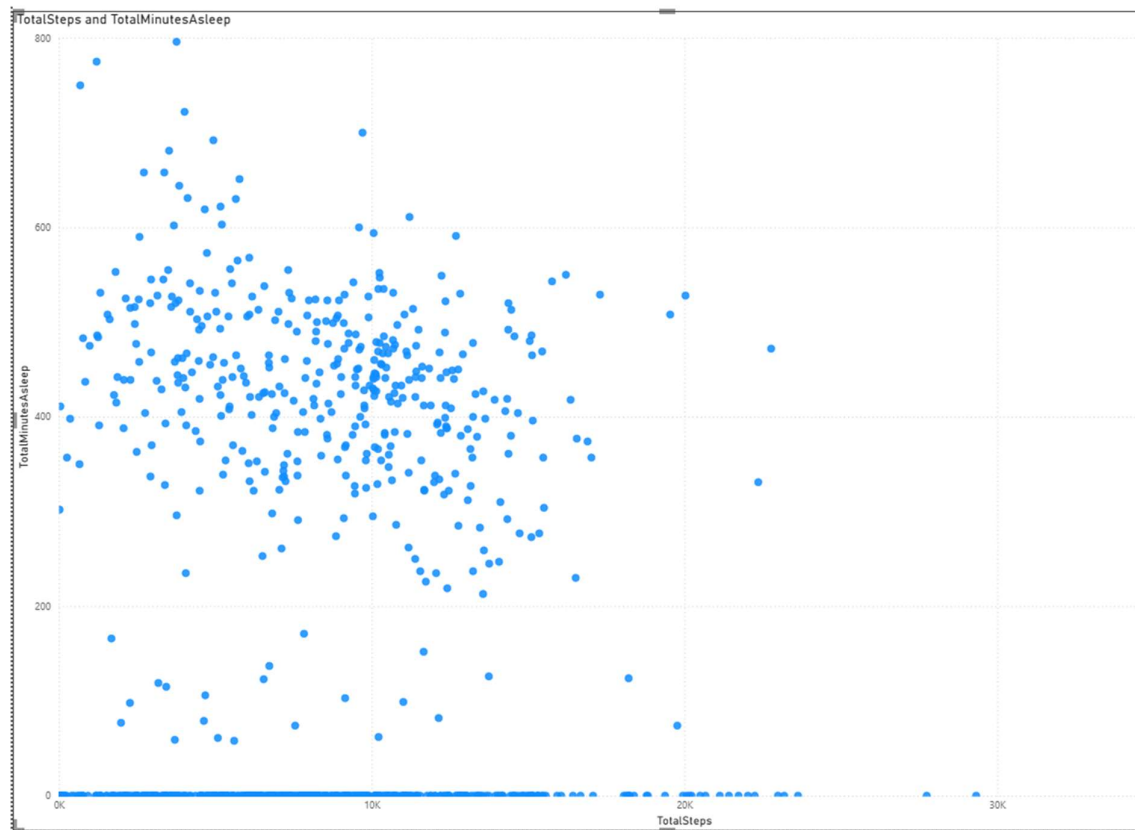
## 8. Sleep vs Activity

Users who have higher daily step counts tend to show better sleep duration.

Insight:

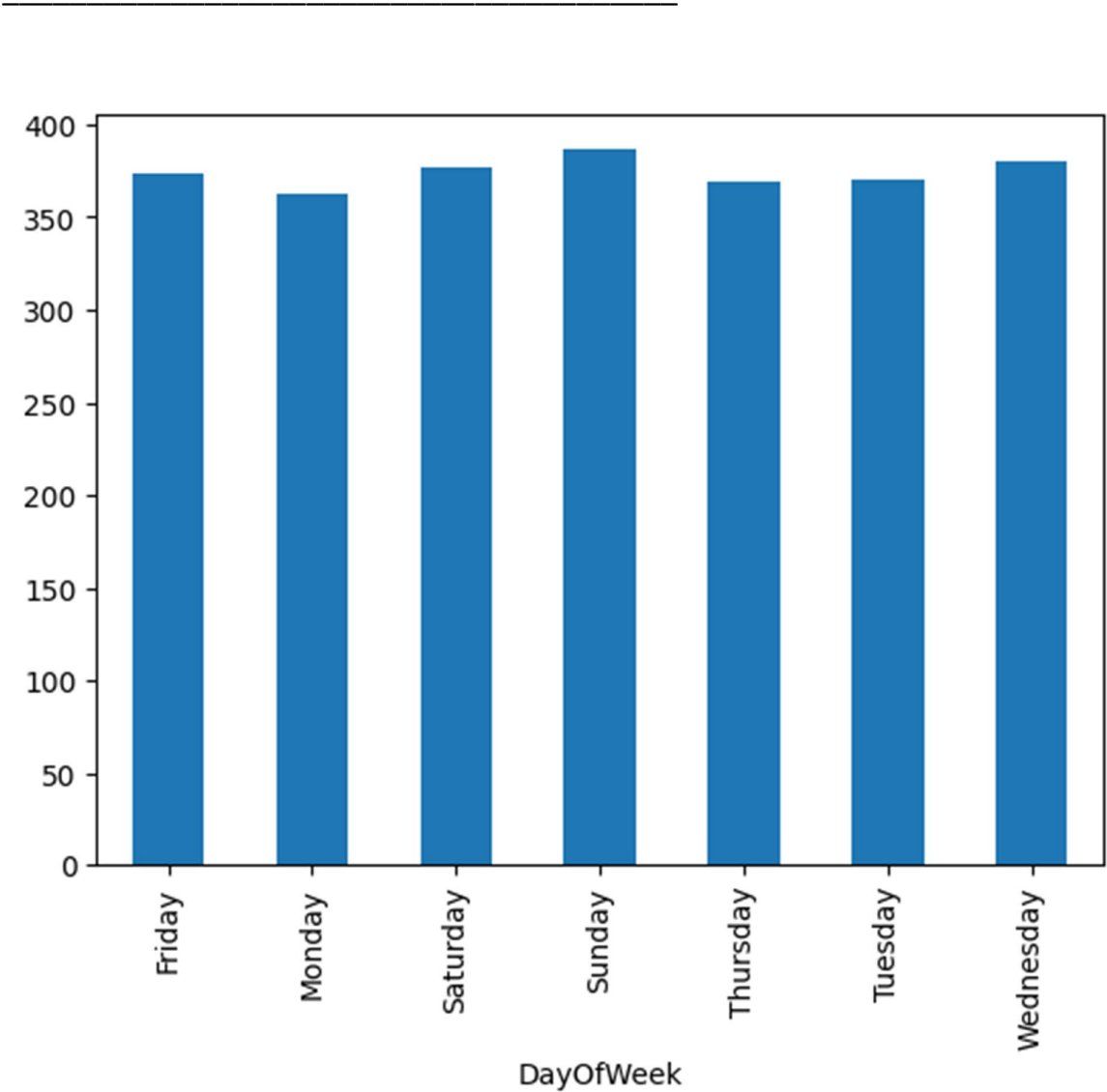
Physical activity may contribute to improved sleep quality.

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## 9. Sleep by Weekday

Sleep duration tends to increase during weekends, indicating users compensate for weekday sleep deficits.



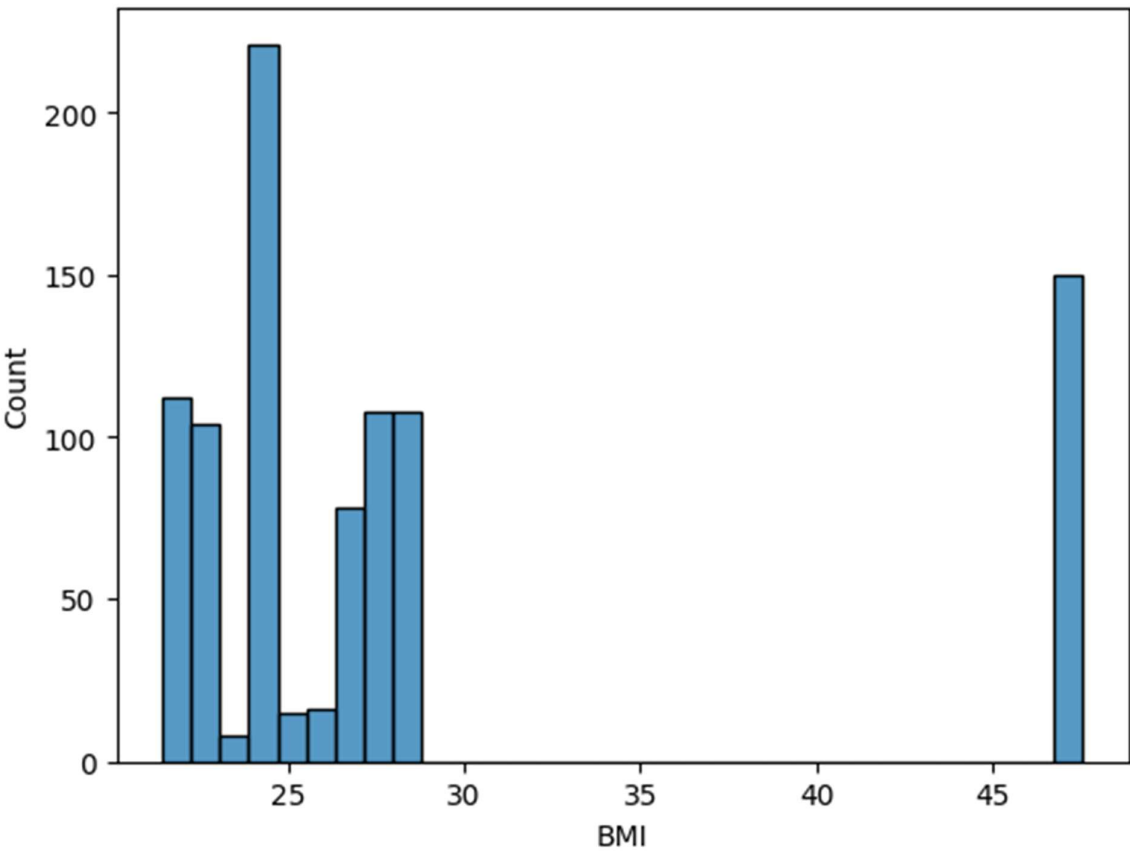
# Weight & BMI Insights

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## 10. BMI Distribution

Most users fall under Normal or Overweight BMI categories.

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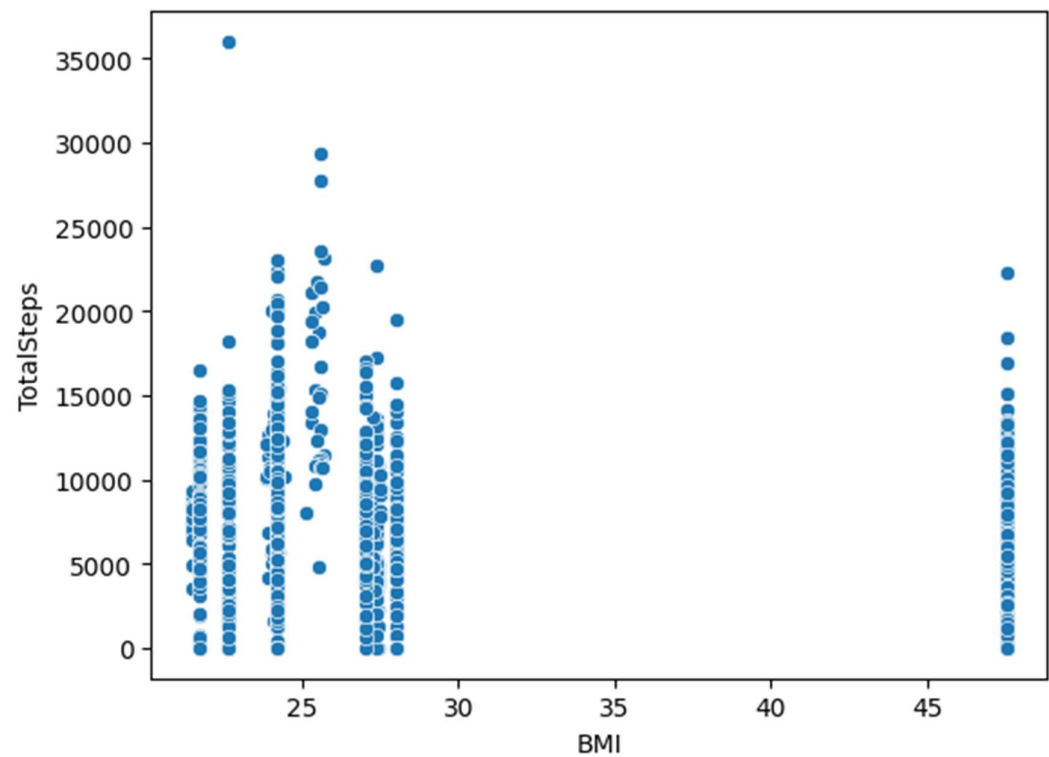


# 11. BMI vs Activity

Users with lower BMI tend to have higher daily step counts.

Insight:

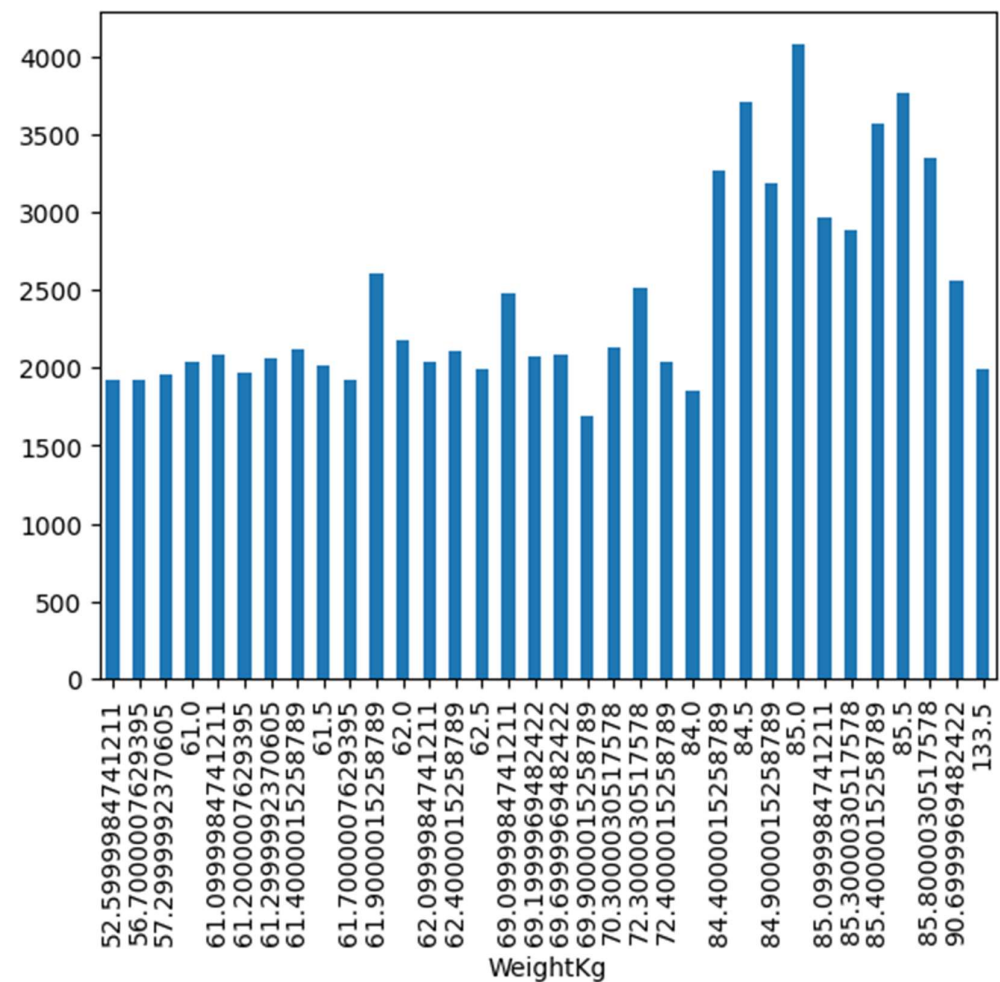
Regular physical activity contributes to healthier body weight.



## 12. Weight vs Calories

Users with higher weight often burn more calories during similar activities, due to higher energy expenditure.

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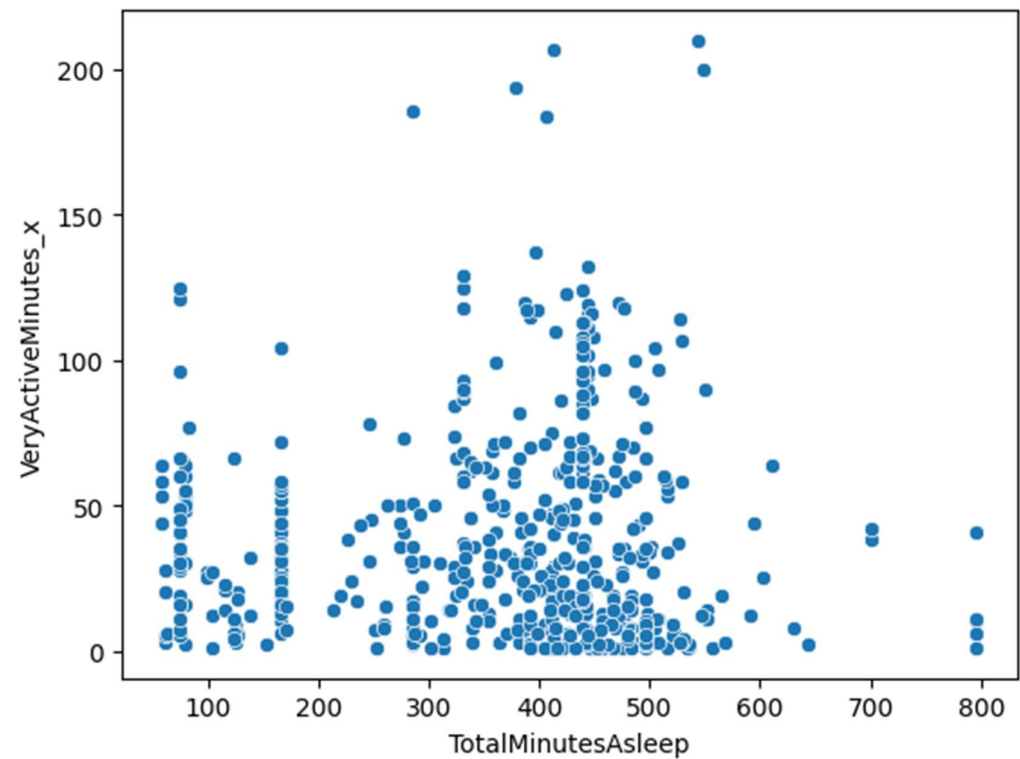
# Combined Health Insights

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## 13. Active Users Sleep Better

Higher physical activity is associated with better sleep efficiency and longer sleep duration.

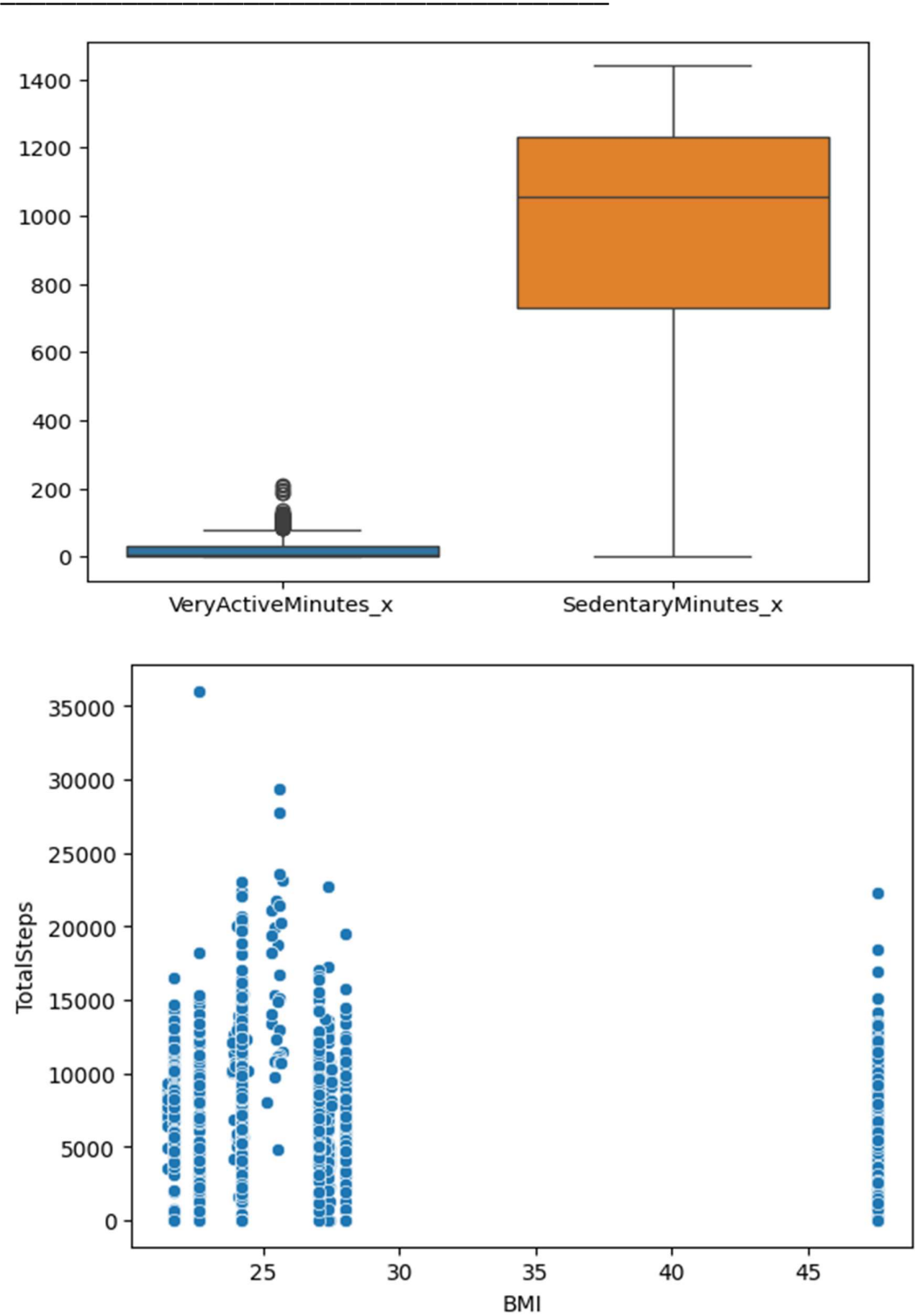
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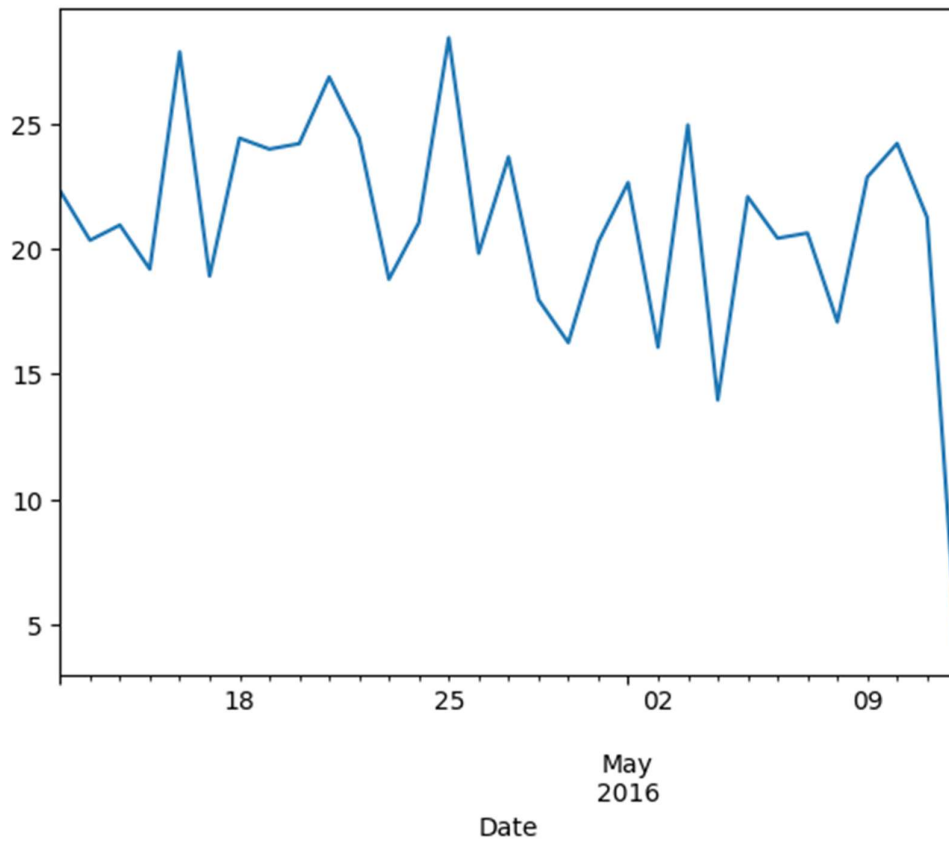
# 14. Sedentary Lifestyle Risk

Users with low steps and high sedentary minutes show signs of higher BMI.



## 15. Peak Engagement Days

Some days show higher activity spikes which may correspond to user motivation events such as workouts or weekends.



# Final Business Recommendation

Based on this analysis, a fitness app could improve engagement by:

- Users can be segmented into:
  - Highly active users (>10,000 steps)
  - Moderately active users (5,000–10,000 steps)
  - Low activity users (<5,000 steps)

Marketing campaigns can be personalized:

- Active users → Promote advanced fitness plans or premium subscriptions
- Moderate users → Encourage step challenges
- Low activity users → Send motivational reminders
- Physical activity contributes to improved sleep quality.

Marketing Strategy Campaign idea:

“Walk more, sleep better.”

This can encourage users to stay active.

- Different BMI groups may require different health goals.

Marketing Strategy Personalized campaigns:

| BMI Category | Campaign Focus                |
|--------------|-------------------------------|
| Underweight  | Nutrition & strength programs |
| Normal       | Maintain healthy lifestyle    |
| Overweight   | Weight loss challenges        |
| Obese        | Beginner fitness programs     |

- Highly active users interact frequently with the app. These users are potential brand promoters.

Introduce:

- Referral programs
- Community challenges
- Social sharing

Users who consistently track activity and health data may be more willing to pay for premium features. Highly engaged users are more likely to convert to paid subscriptions.

Offer premium features like:

- Personalized coaching
- Advanced analytics
- Nutrition tracking